

Global CO₂ Emissions:

Who Emits, How Much, and What's Changing

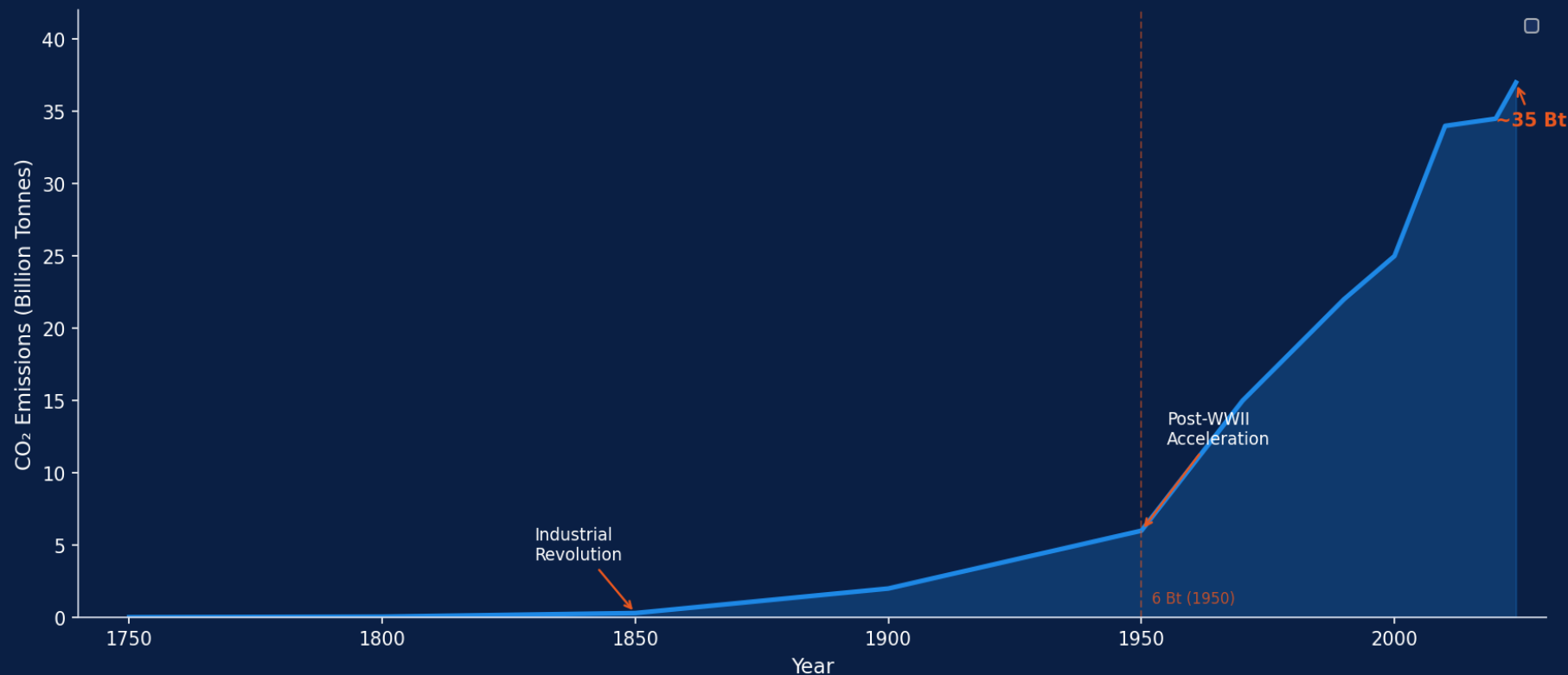
Scope: Fossil fuels & industry · 1750–2024

Data: Global Carbon Budget 2025 via Our World in Data
Hannah Ritchie & Max Roser · ourworldindata.org

Audience: Policy Analysts · Climate Negotiators · Sustainability Researchers

Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

Global CO₂ Emissions: 1750-2024



6 Bt

1950 baseline

Fossil fuel & industrial emissions

~20 Bt

~1990

Near quadrupling in 40 years

35+ Bt

2024

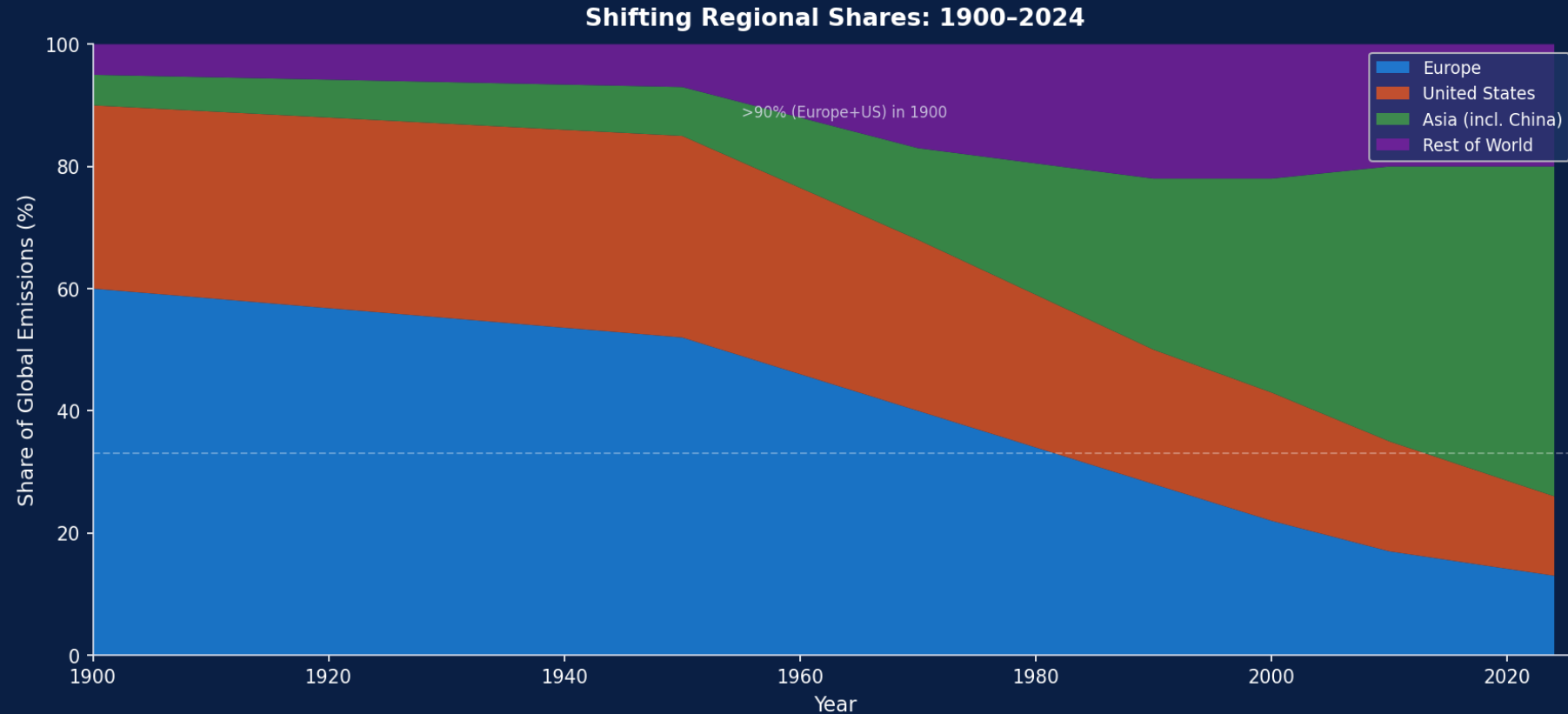
Global total — no peak yet

+5.2×

since 1950

Despite climate commitments

Europe and the US Fell from >90% of Emissions in 1900 to Under One-Third Today



>90%

1900

Europe + US combined

<33%

2024

Same two regions today

~54%

2024

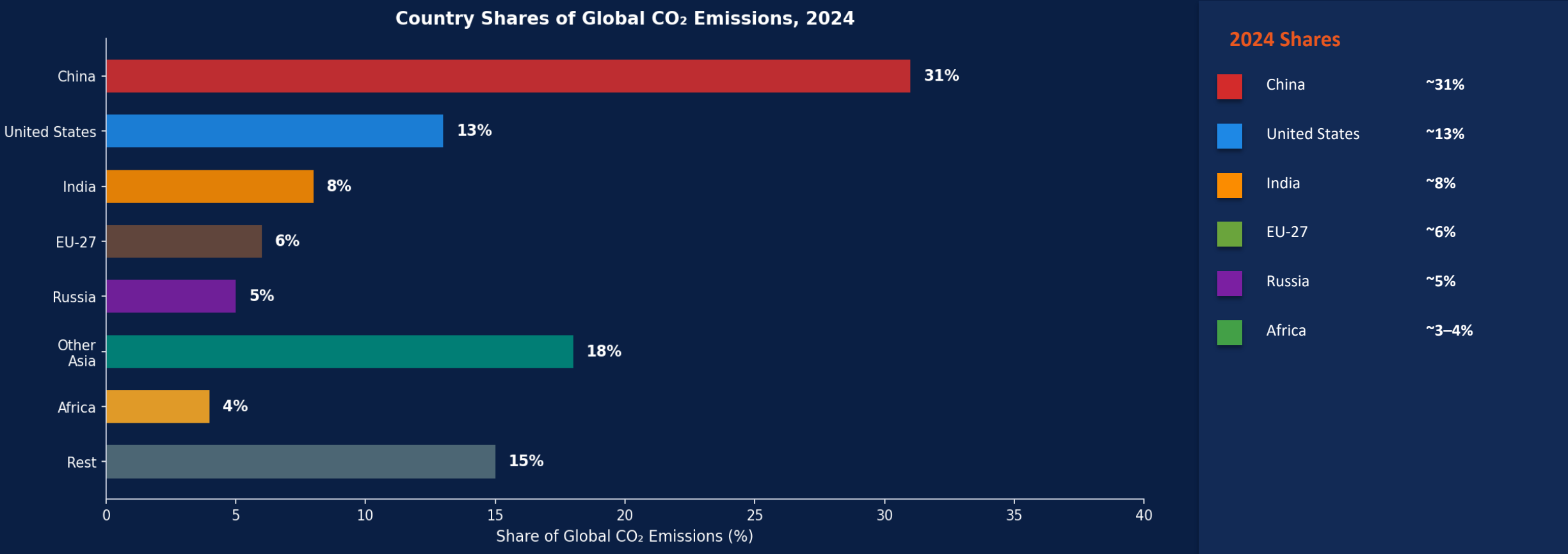
Asia (led by China)

~3x

Shift

In just 75 years

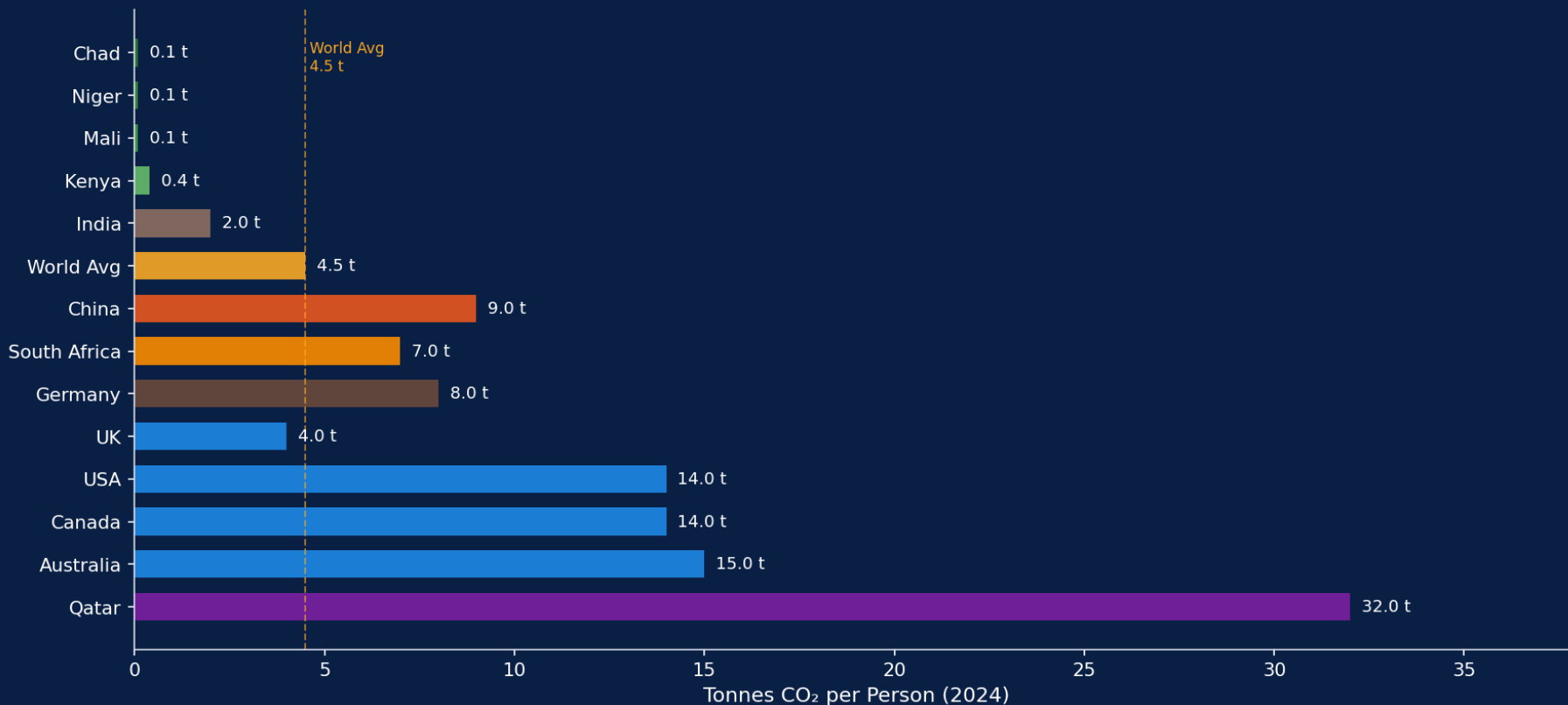
China Now Emits More Than One-Quarter of Global CO₂



Source: Global Carbon Budget 2025, Our World in Data

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada

Per Capita CO₂ Emissions: 150× Gap from Chad to Qatar



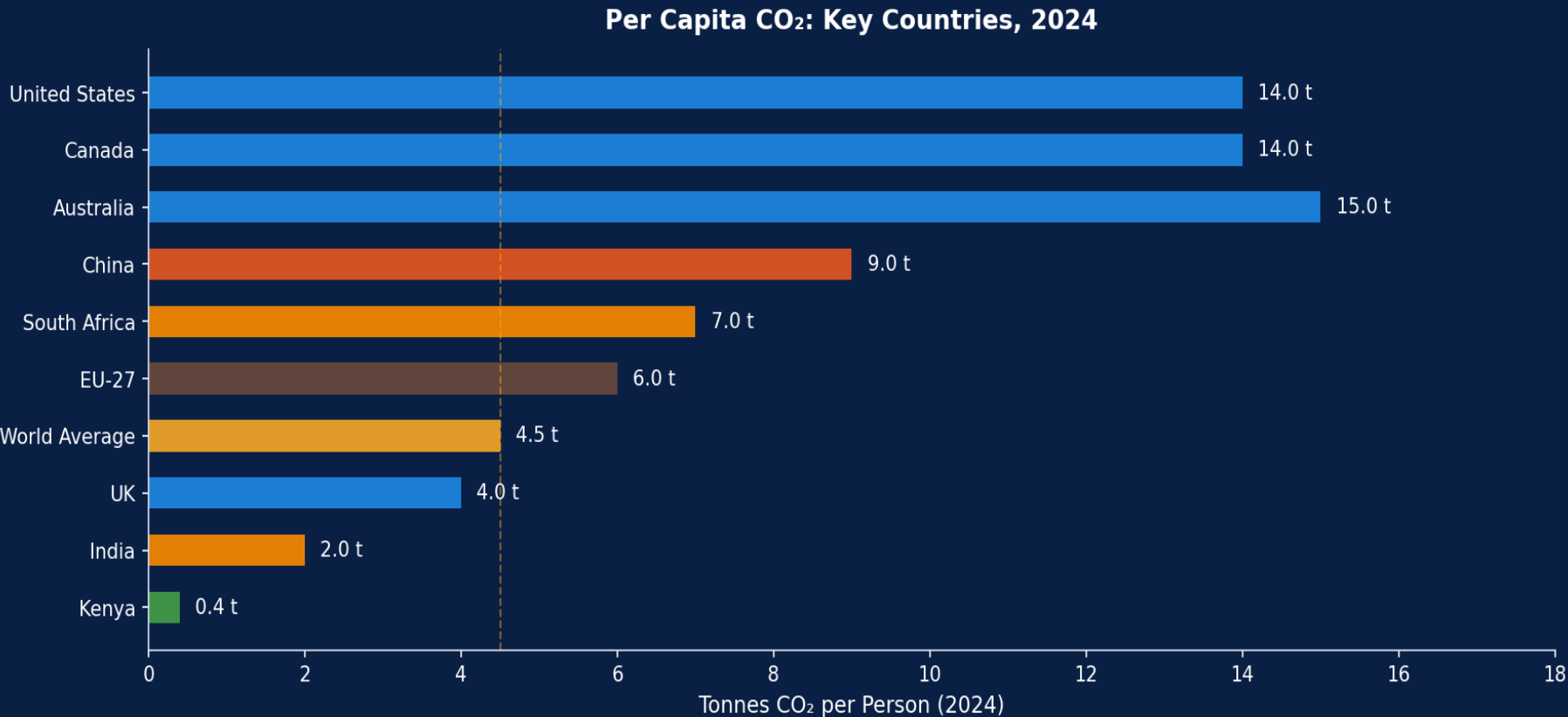
~150×

Gap between highest
and lowest emitting nations

An American emits in
~2 days what a person in
Mali or Niger emits in
a full year.

*France & UK: ~4–5 t
(lower than US despite
similar living standards)*

Per Capita Emissions: ~14 t in the US vs. ~9 t in China and ~2 t in India



Key Insight

China: ~9 t/person

Largest total emitter,
but below US per capita

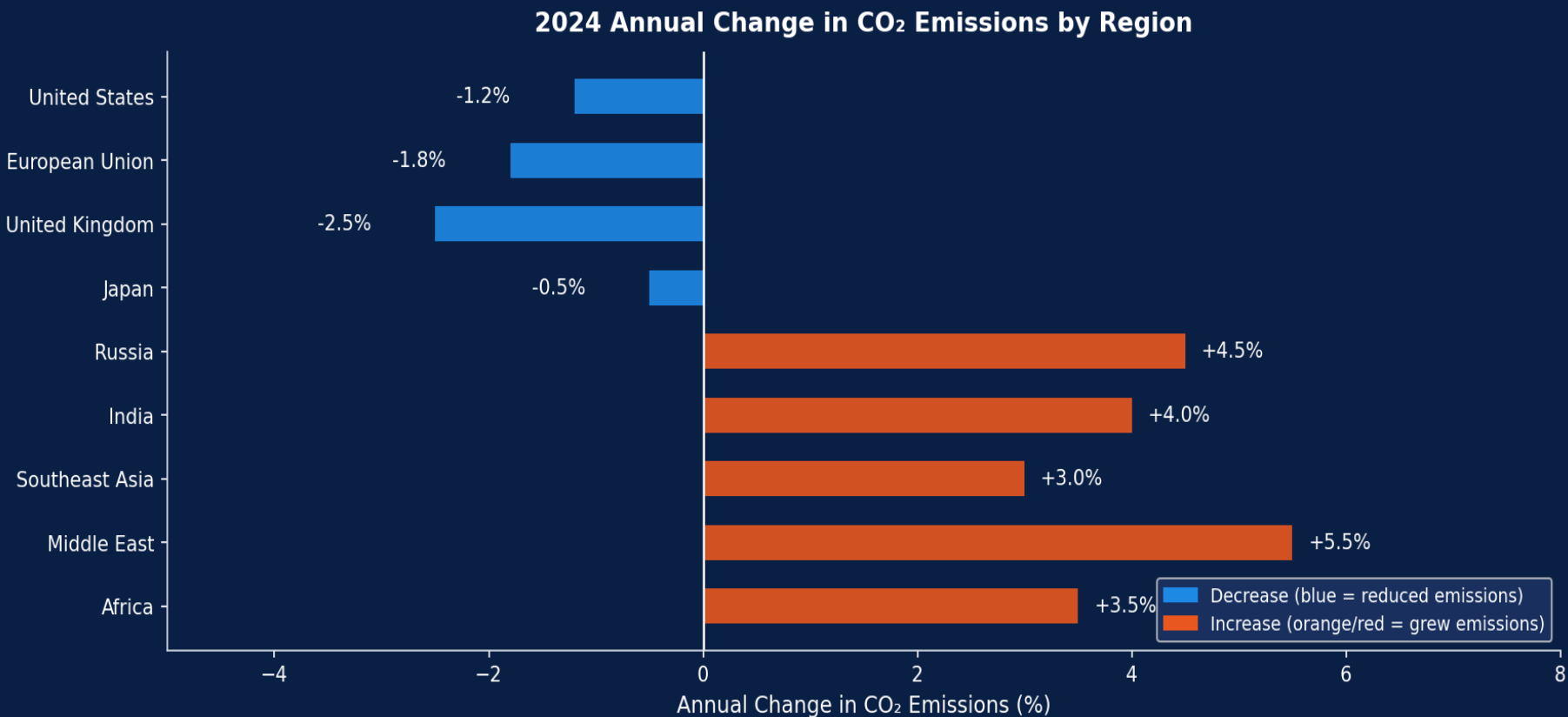
India: ~2 t/person

1.4 billion people,
lowest among major economies

US: ~14 t/person

3× the global avg;
highest among large economies

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew



Declines ↓

United States -1.2%
European Union -1.8%

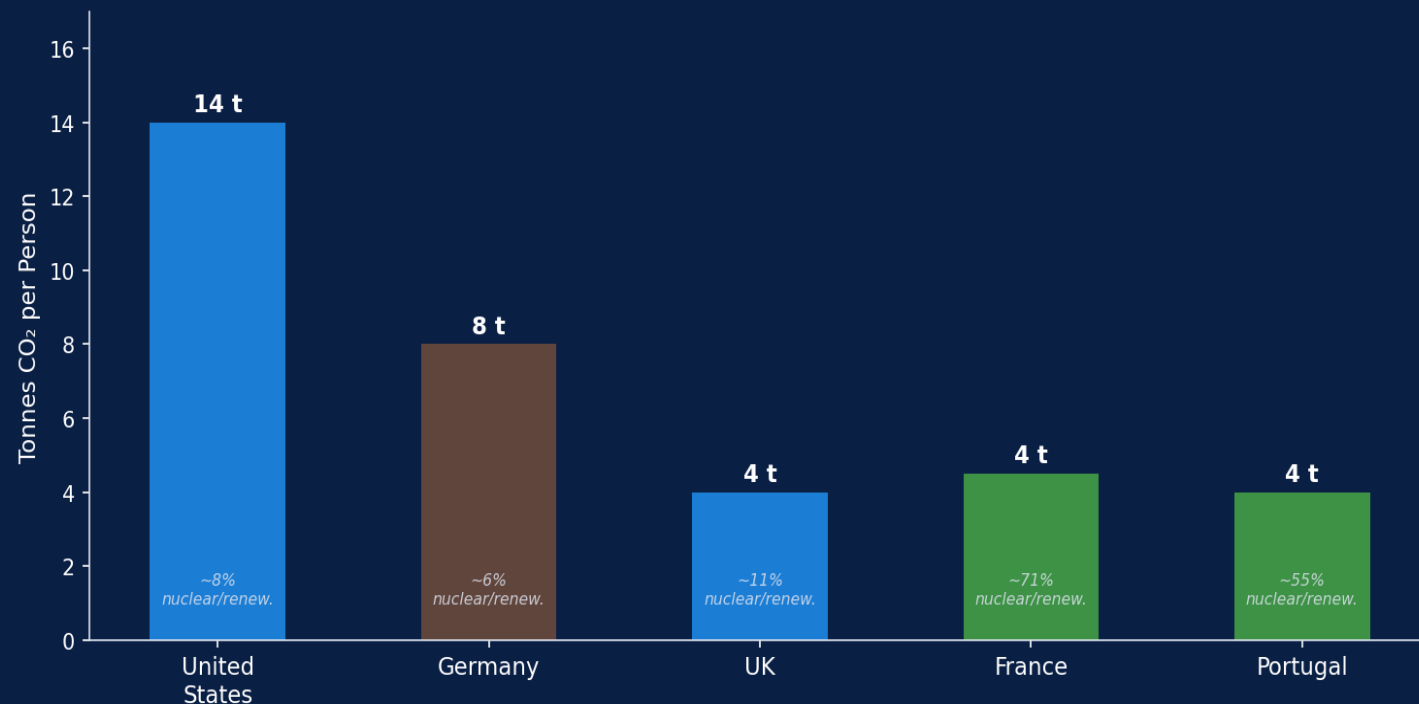
Increases ↑

Russia +4.5%
India +4.0%
Southeast Asia +3.0%

Note: Turkmenistan & Uzbekistan saw >10% increases in 2024.

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US

Per Capita Emissions: Impact of Energy Mix



Why Energy Mix Matters

► Prosperity ≠ High Emissions

France and the UK have living standards comparable to Germany and the US, yet emit 2–3× less per capita.

► Nuclear & Renewables Cut Emissions

France (~71% nuclear) and Portugal (~55% renewable/nuclear) have among the lowest per capita emissions in the wealthy world.

► Fossil Fuels Drive the Gap

Germany and the US rely more heavily on coal and gas for electricity, keeping per capita emissions high despite renewable growth.

► Policy Choices Are Key

The US at ~14 t/person vs. France at ~4.5 t/person shows that energy policy — not wealth alone — shapes emissions.

Key Takeaways

Responsibility Is Shared but Unequal

1 No peak yet

Global CO₂ from fossil fuels exceeds 35 Bt/yr and has not peaked despite slowing growth rates.

2 China is the largest annual emitter — but per capita still below the US

China (~31%) now emits more than the US and EU combined, yet per capita (~9 t) remains well below the US (~14 t).

3 Historical responsibility is dominated by Europe and the US

Both regions together emitted >90% of all CO₂ in 1900 and still account for the vast majority of the cumulative stock.

4 A 150× per capita gap persists between richest and poorest nations

From ~0.1 t/person in Chad/Niger to ~14–15 t/person in the US/Canada/Australia — the largest inequality in the climate crisis.

5 Energy policy can decouple prosperity from emissions

France (~4.5 t) and the UK (~4 t) achieve living standards comparable to Germany (~8 t) and the US (~14 t) through nuclear and renewable electricity.